Embedded Control for Robotics

Lecture # 08 Routh-Hurwitz Stability Criterion

Book: Embedded Digital Control with Microcontrollers
Chapter # 8

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Overview Today's Lecture

- Closed-loop stability
- Stability of Higher Order System
- Routh-Hurwitz Stability Criterion
- Examples
- Quiz # 4 (Written Solution)

Stability

$$G_T(s) = \frac{Y(s)}{R(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$
$$= K \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$$

$$y(t) = K_1 e^{p_1 t} + K_2 e^{p_2 t} + \cdots + K_n e^{p_n t} + y_p(t)$$

If all the exponential terms in the homogeneous solution converge to zero as time goes to infinity, the system is stable.

$$\mathrm{Re}\{p_i\} < 0, i = 1, 2, \cdots, n$$

Stability

- A linear, time-invariant system is stable if the natural response approaches zero as time approaches infinity.

- A linear, time-invariant system is unstable if the natural response grows without bound as time approaches infinity.

- A linear, time-invariant system is marginally stable if the natural response neither decays nor grows but remains constant or oscillates as time approaches infinity

Closed Loop Stability?

- Let us focus on the natural response definitions of stability

- Stable systems have closed-loop transfer functions with poles only on the left half-plane.

- If the closed-loop system poles are on the right half of the s-plane and hence have a positive real part, the system is unstable.

Stability from TF

$G(s)$	Stable	marginally stable	unstable
$\frac{5(s+2)}{(s+1)(s^2+s+1)}$		?	
$\frac{5(-s+2)}{(s+1)(s^2+s+1)}$		?	
$\frac{5}{(s-2)(s^2+3)}$		?	
$\frac{s^2+2}{(s+1)(s^2-s+1)}$		?	
$\frac{5}{(s+2)(s^2+1)^2}$		?	
$\frac{1}{(s^2-1)(s+1)}$		?	

Stability of Higher Order Systems

$$\frac{C_1(s)}{R_1(s)} = \frac{10(s+1)}{s^5 + 3s^3 + 2s + 5}$$

Un-Stable

$$\frac{C_2(s)}{R_2(s)} = \frac{10(s+1)}{s(s^4 + s^3 + 3s^2 + s + 2)}$$

Stable

Routh-Hurwitz Stability Criterion

- It is a method for determining continuous system closed-loop stability.

- The Routh-Hurwitz criterion states that

"the number of roots of the characteristic equation with positive real parts is equal to the **number of changes in sign** of the first column of the Routh array".

Continue…

- This method derives stability information without the need to solve for the closed-loop system poles.
- Using this method, we can tell how many closed-loop system poles are in the left half-plane, in the right half-plane, and on the jw-axis. (Notice that we say how many, not where.)

- The method requires two steps:

1. Generate a data table called a Routh table.
2. interpret the Routh table to tell how many closed-loop system poles are in the LHP, the RHP, and on *the jw-axis*.

Data Table

- The characteristic equation of the nth order continuous system can be write as:

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0$$

- The stability criterion is applied using a Routh table which is defined as;

s^n	a_n	a_{n-2}	a_{n-4}	$\cdots$
s^{n-1}	a_{n-1}	a_{n-3}	a_{n-5}	$\cdots$
$\cdot$	b_1	b_2	b_3	$\cdots$
$\cdot$	c_1	c_2	c_3	$\cdots$
$\cdot$	$\cdots\cdots\cdots\cdots$			

- Where $a_n, a_{n-1}, \ldots, a_0$ are coefficients of the characteristic equation.

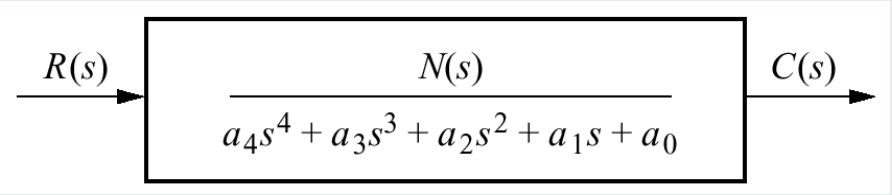
$$b_1 \equiv \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}} \qquad b_2 \equiv \frac{a_{n-1}a_{n-4} - a_n a_{n-5}}{a_{n-1}}$$

$$c_1 \equiv \frac{b_1 a_{n-3} - a_{n-1} b_2}{b_1} \qquad c_2 \equiv \frac{b_1 a_{n-5} - a_{n-1} b_3}{b_1}$$

Continue…

- First label the rows with powers of s from highest power of s down to lowest power of s in a vertical column.
- Next form the first row of the Routh table, using the coefficients of the denominator of the closed-loop transfer function (characteristic equation).
- Start with the coefficient of the highest power and skip every other power of s.
- Now form the second row with the coefficients of the denominator skipped in the previous step.
- The table is continued horizontally and vertically until zeros are obtained.

Generating a basic Routh Table



- Only the first 2 rows of the array are obtained from the characteristic eq. the remaining are calculated as follows;

s^4			
s^3			
s^2			
s^1			
s^0			

Stability: Routh-Hurwitz Method

s^n	1	a_{n-2}	a_{n-4}	
s^{n-1}	a_{n-1}	a_{n-3}	a_{n-5}	
s^{n-2}	b_1	b_2	b_3	
s^{n-3}	c_1	c_2	c_3	$\cdots$
s^{n-4}	d_1	d_2	d_3	
$\vdots$		$\vdots$		
s^0				

$$b_1 = \frac{-\begin{vmatrix} 1 & a_{n-2} \\ a_{n-1} & a_{n-3} \end{vmatrix}}{a_{n-1}} = \frac{a_{n-1}a_{n-2} - a_{n-3}}{a_{n-1}}$$

$$b_2 = \frac{-\begin{vmatrix} 1 & a_{n-4} \\ a_{n-1} & a_{n-5} \end{vmatrix}}{a_{n-1}} = \frac{a_{n-1}a_{n-4} - a_{n-5}}{a_{n-1}}$$

$$c_1 = \frac{-\begin{vmatrix} a_{n-1} & a_{n-3} \\ b_1 & b_2 \end{vmatrix}}{b_1} = \frac{b_1 a_{n-3} - b_2 a_{n-1}}{b_1}$$

$$c_2 = \frac{-\begin{vmatrix} a_{n-1} & a_{n-5} \\ b_1 & b_3 \end{vmatrix}}{b_1} = \frac{b_1 a_{n-5} - b_3 a_{n-1}}{b_1}$$

Continue…

$$A(s) = s^2 + a_1 s + a_0$$

s^2	1	a_0
s^1	a_1	0
s^0	$\dfrac{a_1 a_0 - (1)(0)}{a_1} = a_0$	0

if $a_1 > 0$ and $a_0 < 0 \Rightarrow$ one sign change $\Rightarrow$ 1 RHP pole $\Rightarrow$ unstable

if $a_1 < 0$ and $a_0 < 0 \Rightarrow$ one sign change $\Rightarrow$ 1 RHP pole $\Rightarrow$ unstable

if $a_1 < 0$ and $a_0 > 0 \Rightarrow$ two sign changes $\Rightarrow$ 2 RHP poles $\Rightarrow$ unstable

Continue…

$$A(s) = s^3 + a_2 s^2 + a_1 s + a_0$$

s^3	1	a_1
s^2	a_2	a_0
s^1	$\dfrac{a_2 a_1 - (1)a_0}{a_2} = a_1 - \dfrac{a_0}{a_2}$	0
s^0	a_0	0

if $a_2 < 0$ and $a_0 > 0 \Rightarrow$ two sign changes $\Rightarrow$ 2 RHP poles $\Rightarrow$ unstable

Routh's Stability Condition

- If the closed-loop transfer function has all poles in the left half of the s-plane, the system is stable. Thus, a system is stable if there **are no sign changes in the first column of the Routh table.**

- The Routh-Hurwitz criterion declares that the number of roots of the polynomial that **lie in the right half-plane is equal to the number of sign changes in the first column. Hence, the system is unstable if the poles lie on the right-hand side of the s-plane.**

Example # 1

$$s^5 + 2s^4 + 3s^3 + 4s^2 + 5s + 2 = 0$$

s^5	1	3	5
s^4	2	4	2
s^3			
s^2			
s			
1			

Continue…

$$s^5 + 2s^4 + 3s^3 + 4s^2 + 5s + 2 = 0$$

$$b_1 = \frac{-\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix}}{2} = \frac{2 \cdot 3 - 4}{2} = 1$$

$$b_2 = \frac{-\begin{vmatrix} 1 & 5 \\ 2 & 2 \end{vmatrix}}{2} = \frac{2 \cdot 5 - 2}{2} = 4$$

$$c_1 = \frac{-\begin{vmatrix} 2 & 4 \\ 1 & 4 \end{vmatrix}}{1} = \frac{1 \cdot 4 - 4 \cdot 2}{1} = -4$$

$$c_2 = \frac{-\begin{vmatrix} 2 & 2 \\ 1 & 0 \end{vmatrix}}{1} = \frac{1 \cdot 2 - 0 \cdot 2}{1} = 2$$

$$d_1 = \frac{-\begin{vmatrix} 1 & 4 \\ -4 & 2 \end{vmatrix}}{-4} = \frac{-4 \cdot 4 - 2 \cdot 1}{-4} = \frac{9}{2}$$

$e1=(9/2*2-4*0)/9/2=2$

s^5	1	3	5
s^4	2	4	2
s^3			
s^2			
s			
1			

s^5	1	3	5
s^4	2	4	2
s^3	1	4	
s^2	−4	2	
s	9/2		
1	2		

First-column sequence:
1, 2, 1, −4, 9/2, 21,2,1,−4,9/2,2 → signs:
+, + , + , − , + , +

Number of sign changes = 2 (between $s3 \rightarrow s2$, and $s2 \rightarrow s1$

Two sign changes in the left-most column, the equation has two roots in the right half-plane

Example # 2

$$s^5 + 2s^4 + 4s^3 + 5s^2 + 4s + 2 = 0$$

s^5	1	4	4
s^4	2	5	2
s^3	1.5	3	
s^2	1	2	
s	0	0	
1			

First element of s^3:

$$\frac{(2 \cdot 4) - (1 \cdot 5)}{2} = \frac{8 - 5}{2} = \frac{3}{2}.$$

Second element of s^3:

$$\frac{(2 \cdot 4) - (1 \cdot 2)}{2} = \frac{8 - 2}{2} = \frac{6}{2} = 3.$$

Second element of s^2:

1 2 0

Attempting to compute s^1 entries gives

```
s^1 = 0      0      0
```

Continue…

This shows a repeated/ symmetric polynomial factor; So form the **auxiliary polynomial** from the row *above* (the s^2row).

Auxiliary polynomial (from s^2row `[1, 2]`):

$$A(s) = 1 \cdot s^2 + 2 \cdot s^0 = s^2 + 2.$$

Differentiate $A(s)$:

$$A'(s) = 2s.$$

Use coefficients of $A'(s)$to replace the zero row (arranged for s^1row): coefficients are `[2, 0]`.

So

s^5	1	4	4	
s^4	2	5	2	
s^3	1.5	3		
s^2	1	2	$\rightarrow$	$s^2 + 2$
s	2	0	$\leftarrow$	$\frac{d(s^2+2)}{ds} = 2s$
1	2	0		

Final s^0row Compute s^0using $s^1$and s^2:

first *element of* $s^0 = \frac{(2 \cdot 2) - (1 \cdot 0)}{2} = \frac{4}{2} = 2.$

Stability conclusion

•First-column sequence: $1,\ 2,\ 3/2,\ 1,\ 2,\ 2$— **all positive** (no sign changes).

•**Number of right-half-plane poles** = number of sign changes in the first column = **0.**

•However, the appearance of an entire zero row produced the auxiliary polynomial $A(s)$ $= s^2 + 2$ .The roots of $A(s)$are

$$s = \pm j\sqrt{2},$$

•i.e. two purely **imaginary** poles.

Therefore the system has **no RHP poles** but **two poles on the imaginary axis.** That means the system is **marginally (critically) stable, not asymptotically stable.** (It will sustain undamped oscillations at $\omega = \sqrt{2}$(.

Example # 3

$$s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 2 = 0$$

s^5	1	3	5
s^4	2	6	2
s^3	0	4	
s^2			
s			
1			

s^5	1	3	5
s^4	2	6	2
s^3	ε	4	
s^2	$-8/\varepsilon$	2	
s	4		
1	2		

Special Cases

- **Case-1:** Zero in the first column
  - If first element of a row is zero, division by zero would be required to form the next row.
  - To avoid this phenomenon, zero is replaced by a very small number (say ϵ).

- Sign sequence: +, +, +, -, +, +
- Number of sign changes: 2= First change: from + (ε) to - (-∞) and Second change: from - (-∞) to + (4)
- Conclusion
  - ✅ The system is UNSTABLE
  - ❌ Number of poles in right-half plane: 2
  - ✅ Number of poles in left-half plane: 3 (since it's a 5th-order system: 5 - 2 = 3)
- Final Answer: The system has 2 poles with positive real parts and is therefore unstable.

Example#3

Find the stability of the continues system having the characteristic polynomial of a third order system is given below

$$s^3 + s^2 + 2s + 24$$

- The Routh array is

s^3	1	2
s^2	1	24
s^1	−22	0
s^0	24	0

- Because TWO changes in sign appear in the first column, we find that two roots of the characteristic equation lie in the right hand side of the s-plane. Hence the system is unstable.

Quiz # 4

$$P(s) = 3s^7 + 9s^6 + 6s^5 + 4s^4 + 7s^3 + 8s^2 + 2s + 6$$

1. Draw Routh Table?
2. Determine Stability?

Solution

s^7:	3.000	6.000	7.000	2.000
s^6:	9.000	4.000	8.000	6.000
s^5:	4.667	4.333	0	
s^4:	-4.355	8.000	6.000	
s^3:	12.906	6.429	0	
s^2:	10.167	6.000		
s^1:	-1.188	0		
s^0:	6.000			

- **Four roots** of the characteristic equation lie in the right half-plane (rhp), and **three roots** lie of the characteristic equation in the left half-plane (lhp). Therefore the **system is unstable**.